

Introduction to web programming

Project work

Lappeenranta-Lahti University of Technology LUT

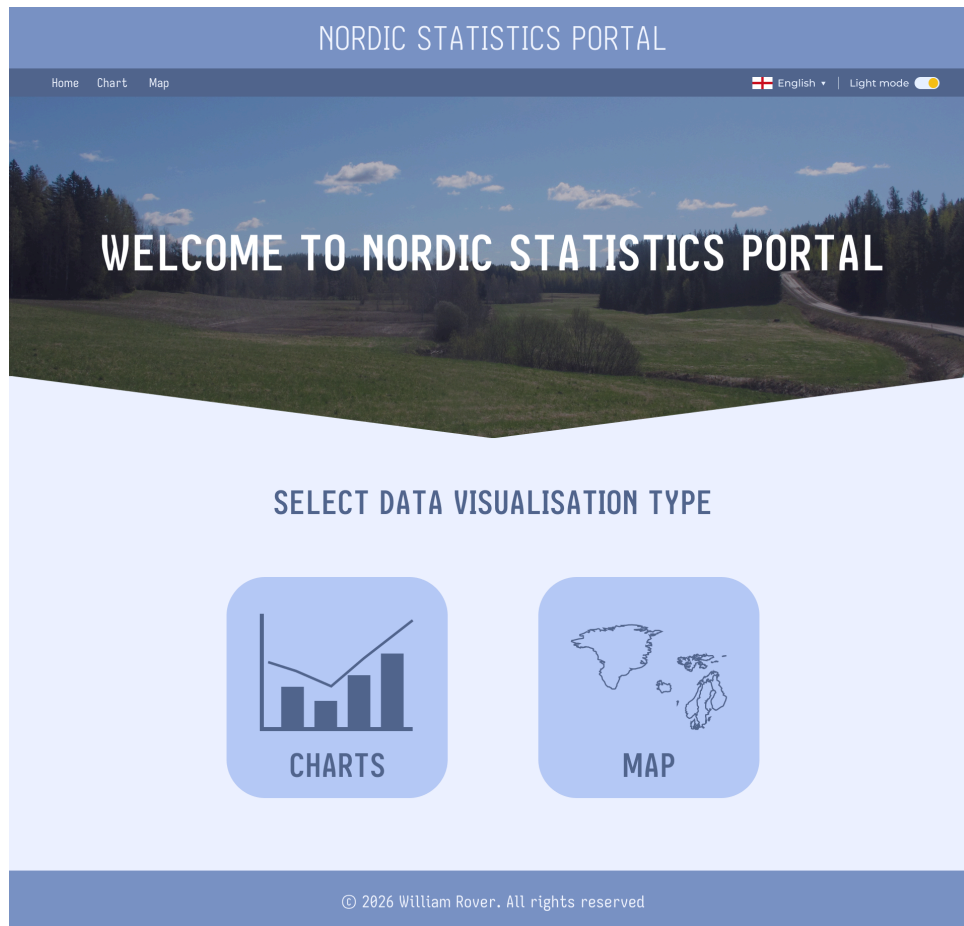
Software Engineering

Nguyen Minh Thuan Bui, 003043417

Introduction to web programming

Summer 2026

NORDIC STATISTICS PORTAL



Site image

Link: nordic-statistics-portal.roverwilliam4567.workers.dev

Nordic Statistics Portal is created to fetch data within the Nordic region, and give users ability to manipulate with the data. For example: Create visualisation, predict data or save as SVG/PNG.

User can choose to visualize data either as a Chart or a Map.

Although the website is responsive on both desktop and mobile environments, it is recommended to use on desktop environment since if there are too many datasets or parameters, the visualisation becomes unreadable.

API USED

All data are fetched from nordicstatistics.org

- POPU04: Marriages and divorces by reporting country, type and time
[https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Family structure/POPU04.px](https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Family%20structure/POPU04.px)
- POPU01: Population 1 January by unit, reporting country, age, sex and time
[https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Population size/POPU01.px](https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Population%20size/POPU01.px)
- DEAT01: Deaths by reporting country, age, sex and time
[https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Population change/DEAT01.px](https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Population%20change/DEAT01.px)
- CHIL01: Live births by reporting country, age of mother and time
<https://pxweb.nordicstatistics.org:443/api/v1/en/NordicStatistics/Demography/Fertility/CHIL01.px>

FEATURES IMPLEMENTED

Feature	Max points	Did ?	Where
Well written PDF report	3	Y	this
No report	-30	N	
Application is responsive and can be used on both desktop and mobile environment	4	Y	
Application is not responsive	-2	N	
Application works on Firefox, Safari, Edge and Chrome	3	Y	
The application has clear directory structure and everything is organized well	2	Y	
Application does not work	-30	N	
CSS, JavaScript and HTML are all in the same file	-5	N	
Inappropriate content, including hate speech -related memes and other trash	-100	N	
Drag'n'drop new data to charts/maps (there are different data items on screen that user can drag and drop as she wishes)	4	Y	Chart page
The application show relevant data on a map and user has chance to change the data	3	Y	In Map page, user can change general parameters or data item
The application show relevant data on a chart and user has a chance to change the data	3	Y	In Chart page, user can change general parameters or data item
User is able to switch between different layers of data on map	2	Y	The 'Showing data' section in Map page
By clicking the map user has an option to get to additional charts covering that area	4	Y	It will be cluttering so I only show chart for the current

			selected data item. Detailed charts will be its own page
There are more than one item of data available (e.g. elections data, employment rate and number of residents) – this means that there are two API calls made	3	Y	POPU04, POPU01, DEAT01, CHIL01
There are more than two items of data available (e.g. elections data, employment rate and number of residents) – this means that there are three API calls made	2	Y	POPU04, POPU01, DEAT01, CHIL01
Data is combined and merged to generate new data, which is then visualized	3	Y	Drag'n'drop data into the area, then click generate it
Users can define what should be done to different data items (e.g. values are added, multiplied together etc. before visualization)	2	Y	Add predict data for next year, choose bar or line graph
Able to download the visualization as a PNG (or SVG) image	2	Y	In both pages
Multiple language support	2	Y	Support fi, en and se (will mostly use google translate)
Able to switch light/dark theme	1	Y	On the navbar

AI USAGE

AI assistance (Gemini) was used for technical understanding and assists with code debugging.